

The empirical recurrence for OEIS A300933, recovered from its tail

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Abstract

OEIS A300933 records “Empirical recurrence of order 95” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 110$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 97$. Its last coefficient is $c_{95} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-95 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300933 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 1, 2, 4 or 6 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 50, 143, 505, 1534, 4781, 15292, 48838, ...

The entry states:

Empirical recurrence of order 95 (see link above)

As of the “Last modified” line on the live entry (Mar 16 2018 08:54:26, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 110$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 110$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 95: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 95.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 220$ exact terms from $n = 2$ on, it returns order 95 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{95} c_i a(n-i),$$

c_1	13	c_{25}	−51880	c_{49}	−33301240	c_{73}	996800
c_2	−74	c_{26}	455718	c_{50}	−30225446	c_{74}	−652386
c_3	258	c_{27}	−491876	c_{51}	1230846	c_{75}	−1140503
c_4	−697	c_{28}	1278588	c_{52}	1343776	c_{76}	−1012067
c_5	1763	c_{29}	1104888	c_{53}	−28113144	c_{77}	−639881
c_6	−4202	c_{30}	−125385	c_{54}	−36248004	c_{78}	−187105
c_7	8710	c_{31}	2435572	c_{55}	−40571835	c_{79}	36096
c_8	−16130	c_{32}	−4188609	c_{56}	−2363851	c_{80}	95036
c_9	28683	c_{33}	−2838016	c_{57}	55191921	c_{81}	82723
c_{10}	−47020	c_{34}	−4464026	c_{58}	84748726	c_{82}	30603
c_{11}	66213	c_{35}	−7637077	c_{59}	89873144	c_{83}	1974
c_{12}	−82704	c_{36}	4876241	c_{60}	54268237	c_{84}	−5567
c_{13}	92811	c_{37}	−1515639	c_{61}	3666109	c_{85}	−5218
c_{14}	−74411	c_{38}	5391734	c_{62}	−30338089	c_{86}	−986
c_{15}	14029	c_{39}	5031065	c_{63}	−48631479	c_{87}	372
c_{16}	69699	c_{40}	−7120195	c_{64}	−44065544	c_{88}	467
c_{17}	−177284	c_{41}	8990421	c_{65}	−27697582	c_{89}	337
c_{18}	289649	c_{42}	3758583	c_{66}	−12854323	c_{90}	5
c_{19}	−314185	c_{43}	20843057	c_{67}	254736	c_{91}	−39
c_{20}	203266	c_{44}	35062370	c_{68}	6875024	c_{92}	−18
c_{21}	−21215	c_{45}	15005748	c_{69}	8330086	c_{93}	−9
c_{22}	−225387	c_{46}	12284114	c_{70}	7975215	c_{94}	1
c_{23}	404342	c_{47}	−27195709	c_{71}	5614315	c_{95}	1
c_{24}	−417411	c_{48}	−48672756	c_{72}	3111035		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 97$, and it is the recurrence OEIS A300933 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{95} - c_1x^{95-1} - \dots - c_{95}$. Its constant term is $-c_{95} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 95 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 95, so $\deg q = 95$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 95, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 95, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-95 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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